The Schwartz–Smith Model for the Front-Month and 12-Month Futures

1. Goal and Data

We observe two futures prices each day:

- the **front-month** future, with time to maturity T_1 (e.g. $T_1 = 1/12$ year). It trades every day;
- the **12-month** future, with time to maturity $T_2 = 1$ year. It trades only on some days.

We treat T_1 and T_2 as fixed (constant maturity). Days are indexed by $t = 1, 2, \ldots$, and Δt is the length of one day in years ($\Delta t = 1/252$). The goal is an estimate of the 12-month price on *every* day, including days when it does not trade, together with a measure of how uncertain that estimate is.

2. The Model in Continuous Time

Let S_t be the spot price. Schwartz and Smith (2000) write

$$\ln S_t = \chi_t + \xi_t,$$

where χ_t is the short-term deviation, which mean-reverts to zero, and ξ_t is the long-term equilibrium level, which follows a Brownian motion with drift:

$$d\chi_t = -\kappa\,\chi_t\,dt + \sigma_\chi\,dz_\chi, \qquad d\xi_t = \mu_\xi\,dt + \sigma_\xi\,dz_\xi, \qquad dz_\chi\,dz_\xi = \rho\,dt.$$

$\kappa > 0$	speed of mean reversion of χ_t (a half-life of $\ln 2/\kappa$ years)
$\sigma_\chi,\ \sigma_\xi > 0$	volatilities of the short-term and long-term factors
$\rho \in (-1, 1)$	correlation between the two Brownian motions z_χ and z_ξ
μ_ξ	drift of the long-term factor
$\lambda_\chi,\ \lambda_\xi$	risk premia for the two factors; we write $\mu_\xi^* = \mu_\xi - \lambda_\xi$

Futures prices. Let $F_t(T)$ be the price on day t of a future with time to maturity T. Schwartz and Smith show that

$$\ln F_t(T) = e^{-\kappa T}\chi_t + \xi_t + A(T),$$

$$A(T) = \mu_\xi^* T - \frac{\lambda_\chi}{\kappa}\big(1 - e^{-\kappa T}\big) + \frac{1}{2}\Big[\frac{\sigma_\chi^2}{2\kappa}\big(1 - e^{-2\kappa T}\big) + \sigma_\xi^2 T + \frac{2\rho\sigma_\chi\sigma_\xi}{\kappa}\big(1 - e^{-\kappa T}\big)\Big].$$

The short-term factor's effect dies out with maturity through the weight $e^{-\kappa T}$. The long-term factor moves all maturities one-for-one. This is why the front month tells us something about the 12-month price.

3. State-Space Form

State vector. $x_t = \begin{pmatrix} \chi_t \\ \xi_t \end{pmatrix}$ is a 2×1 vector that we never observe directly.

Transition equation (how x_t moves from one day to the next):

$$x_t = c + G\,x_{t-1} + w_t, \qquad w_t \sim N(0, W),$$

with c a 2×1 vector, and G and W both 2×2 matrices:

$$c = \begin{pmatrix} 0 \\ \mu_\xi \Delta t \end{pmatrix}, \qquad G = \begin{pmatrix} e^{-\kappa\Delta t} & 0 \\ 0 & 1 \end{pmatrix}, \qquad W = \begin{pmatrix} \dfrac{\sigma_\chi^2}{2\kappa}\big(1 - e^{-2\kappa\Delta t}\big) & \dfrac{\rho\sigma_\chi\sigma_\xi}{\kappa}\big(1 - e^{-\kappa\Delta t}\big) \\ \dfrac{\rho\sigma_\chi\sigma_\xi}{\kappa}\big(1 - e^{-\kappa\Delta t}\big) & \sigma_\xi^2\,\Delta t \end{pmatrix}.$$

These are exact (not approximations) for the continuous-time model of Section 2.

Measurement equation (how the observed prices relate to x_t). On a day when both contracts trade, the observation is the 2×1 vector of log prices:

$$y_t = \begin{pmatrix} \ln F_t(T_1) \\ \ln F_t(T_2) \end{pmatrix} = d + B\, x_t + v_t, \qquad v_t \sim N(0, V),$$

$$d = \begin{pmatrix} A(T_1) \\ A(T_2) \end{pmatrix} \ (2 \times 1), \qquad B = \begin{pmatrix} e^{-\kappa T_1} & 1 \\ e^{-\kappa T_2} & 1 \end{pmatrix} \ (2 \times 2), \qquad V = \begin{pmatrix} s_1^2 & 0 \\ 0 & s_2^2 \end{pmatrix} \ (2 \times 2).$$

Row i of the equation is the futures formula of Section 2 for maturity T_i, plus a pricing error with standard deviation s_i. The errors v_t are independent of w_t.

Days when the 12-month contract does not trade. On such days we drop the second row. Then $y_t = \ln F_t(T_1)$ is a scalar, $d = A(T_1)$, $B = \begin{pmatrix} e^{-\kappa T_1} & 1 \end{pmatrix}$ is 1×2, and $V = s_1^2$. Nothing else changes. A stale 12-month price is never used as data.

4. The Kalman Filter

Write m_t (2×1) and P_t (2×2) for the mean and covariance of x_t given all prices up to and including day t, and $m_{t|t-1}$, $P_{t|t-1}$ for the same quantities given prices only up to day $t-1$. I is the 2×2 identity matrix, and $^\top$ denotes transpose. Start from $m_0 = \begin{pmatrix} 0, & \ln F_1(T_1) - A(T_1) \end{pmatrix}^\top$ and $P_0 = \operatorname{diag}\begin{pmatrix} \sigma_\chi^2/2\kappa, & 1 \end{pmatrix}$. Then, for each day t:

1. **Predict the state:** $m_{t|t-1} = c + G\, m_{t-1}, \quad P_{t|t-1} = G\, P_{t-1}\, G^\top + W.$
2. **Prediction error of the observed prices:** $\nu_t = y_t - (d + B\, m_{t|t-1})$, with covariance $\Sigma_t = B\, P_{t|t-1}\, B^\top + V.$
3. **Update the state:** $K_t = P_{t|t-1} B^\top \Sigma_t^{-1}, \quad m_t = m_{t|t-1} + K_t\, \nu_t, \quad P_t = (I - K_t B)\, P_{t|t-1}.$

On two-price days ν_t is 2×1, Σ_t is 2×2 and K_t is 2×2. On front-only days ν_t and Σ_t are scalars, and K_t is 2×1.

5. Estimation and Output

Parameters. There are nine: $\theta = (\kappa, \sigma_\chi, \sigma_\xi, \rho, \mu_\xi, \lambda_\chi, \mu_\xi^*, s_1, s_2)$. They are estimated by maximizing the log-likelihood that the filter produces,

$$\log L(\theta) = -\frac{1}{2} \sum_t \Big[n_t \ln(2\pi) + \ln \det \Sigma_t + \nu_t^\top \Sigma_t^{-1} \nu_t \Big],$$

where $n_t \in \{1, 2\}$ is the number of prices observed on day t.

Estimated 12-month price. Let $b_2 = \begin{pmatrix} e^{-\kappa T_2} & 1 \end{pmatrix}$ be the second row of B. On every day t, whether or not the contract traded,

$$\widehat{\ln F_t(T_2)} = A(T_2) + b_2\, m_t, \qquad \text{variance} = b_2\, P_t\, b_2^\top.$$

On a front-only day, this estimate moves by $g_t\, \nu_t$, where ν_t is the front-month prediction error and

$$g_t = \frac{b_2\, P_{t|t-1}\, b_1^\top}{b_1\, P_{t|t-1}\, b_1^\top + s_1^2}, \qquad b_1 = \begin{pmatrix} e^{-\kappa T_1} & 1 \end{pmatrix}.$$

g_t is the regression coefficient of the 12-month price on the front-month price implied by the model. The variance $b_2 P_t b_2^\top$ grows on consecutive front-only days and falls back whenever the 12-month contract trades. This tells you how far to trust the estimate.

Reference. Schwartz, E. and J. E. Smith (2000), "Short-term variations and long-term dynamics in commodity prices," *Management Science* 46(7), 893–911.